

- Another benefit is that one can use strongly typed integrated language, aka Language Integrated Query (LINQ), to fire the DB queries.
- It also implements 'unit of work' out-of-the-box to help track changes and concurrency.
- Finally, it's data-binding friendly; this means that if you are using WPF or Winforms data binding frameworks, it is friendly with these.

Steps to create 'Hello World' using EF 5.0 Core

Prerequisites

- EF Core is a .NET Standard 2.0 library. So, it requires a .NET implementation that supports .NET Standard 2.0 to run. For example, you can use EF Core to develop apps that target .NET Core. Building .NET Core apps requires the .NET Core SDK.
- IDE:** You can use any development environment like Visual Studio, Visual Studio for Mac, or Visual Studio Code. You can use EF Core to develop applications on Windows using Visual Studio. The latest version of Visual Studio is recommended.
- EF Core can run on other .NET implementations like Xamarin and .NET Native, too. But in practice, those implementations have runtime limitations that may affect how well

EF Core	2.1 and 3.1	5
.NET Standard	2	2.1
.NET Core	2	3
.NET Framework(1)	4.7.2	(not supported)
Mono	5.4	6.4
Xamarin.iOS(2)	10.14	12.16
Xamarin.Mac(2)	3.8	5.16
Xamarin.Android(2)	8	10
UWP(3)	10.0.16299	TBD
Unity(4)	2018.1	TBD

Table 1: The .NET implementations supported by EF Core

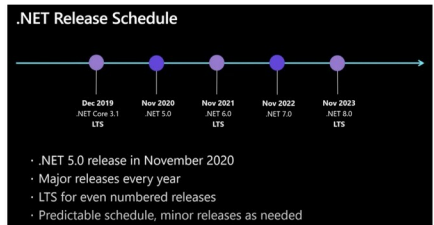


Figure 2: .NET 5 release timeline

Supported versions				
Version	Status	Latest release	Latest release date	End of support
.NET 5.0 (recommended)	Current	5.0.0	2020-11-10	
.NET Core 3.1	LTS	3.1.10	2020-11-10	2022-12-03
.NET Core 2.1	LTS	2.1.23	2020-10-13	2021-08-21

Figure 3: Versions supported by .NET 5

EF Core works on your app. Table 1 shows various .NET implementations supported by EF Core.

How to get EF Core runtime

To add EF Core to an application, we need to install the required NuGet package of the database provider to start using it. Let's say, if we are building an ASP.NET Core application that has SQL Server as a database, then

the EF Core runtime of SQL Server is what is required, which can be installed via NuGet.

We can use the .NET Core command line interface (CLI) or the Visual Studio Package Manager Dialog, or the Visual Studio Package Manager Console to get the required packages.

Assuming you have installed .NET Core runtime, go ahead and type the command given below in your .NET Core CLI command window:

```
D:\EF\myfirstefcore> dotnet add package Microsoft.EntityFrameworkCore.SqlServer
```

Get the .NET Core CLI tools

dotnet ef must be installed as a global or local tool. Most developers prefer installing *dotnet ef* as a global tool using the following command:

```
D:\EF\myfirstefcore> dotnet tool install --global dotnet-ef
```